

Acidity / Basicity
 Acid? Lewis $\Rightarrow e^-$ acceptor

Bronsted $\Rightarrow H^+$ donor

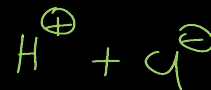
Ionized



Strong acid

HCl
 (weak acid)

gas phase
 $\xrightarrow{\times}$

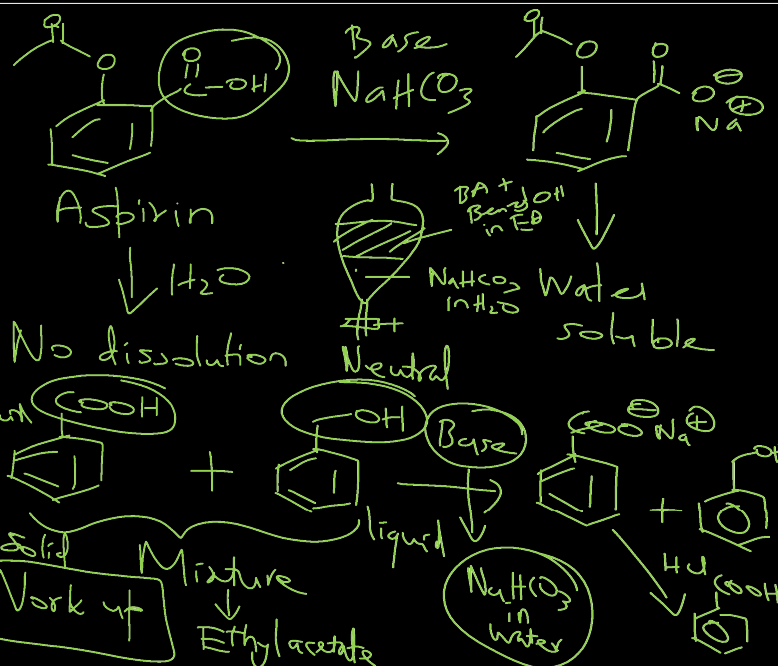


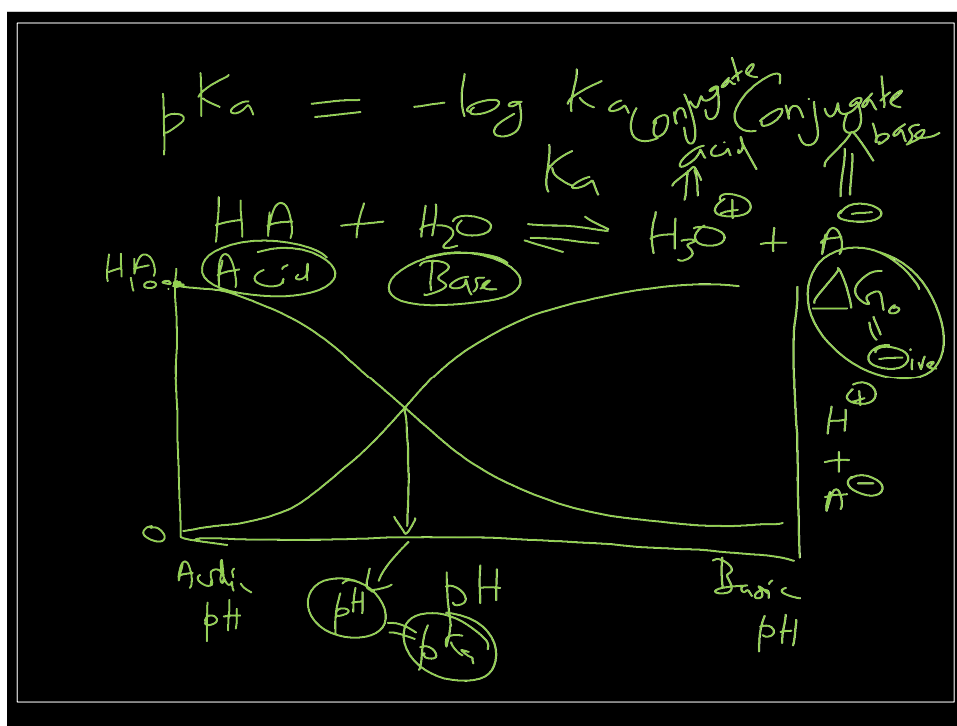
1
 Acidic

7

14
 Basic

$$pH = -\log[H_3O^+]$$





$HI \rightarrow H^+ + I^-$

$HCl \rightarrow H^+ + Cl^-$

① Size of I⁻/Cl⁻

② Bond strength

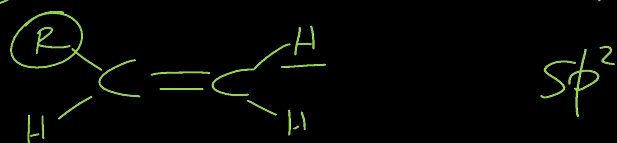
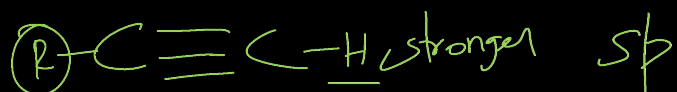
$H_2SO_3 \rightleftharpoons HSO_3^- + H^+$

$H_2SO_4 \rightleftharpoons HSO_4^- + H^+$

③ Resonance

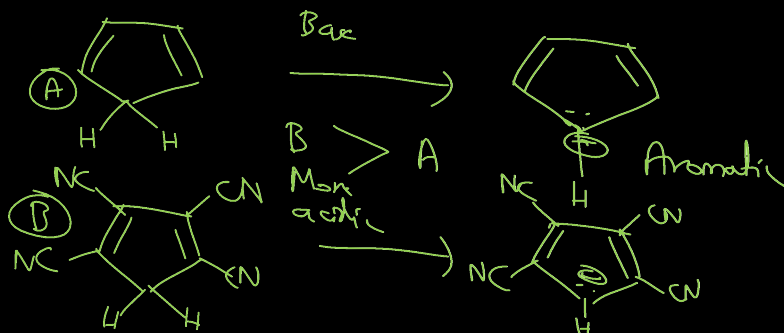
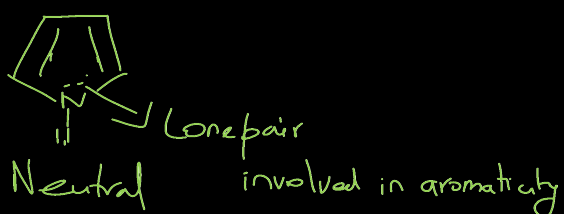
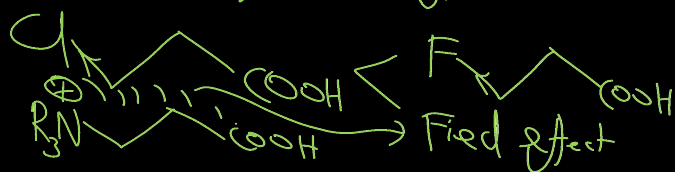
Approximate

Acid	pK _a	Conjugate base
HI	ca. -10	I ⁻
HCl	ca. -7	Cl ⁻
H ₂ SO ₄	ca. -3	HSO ₄ ⁻
HSO ₄ ⁻	2.0	SO ₄ ²⁻
CH ₃ COOH	4.8	CH ₃ COO ⁻
H ₂ S	7.0	HS ⁻
NH ₄ ⁺	9.2	NH ₃
C ₆ H ₅ OH	10.0	C ₆ H ₅ O ⁻
CH ₃ OH	15.5	CH ₃ O ⁻
CH ₃ COCH ₃	20.0	CH ₃ COCH ₂ ⁻
CH ₃ C≡CH	25	CH ₃ C≡C ⁻
NH ₃	33	NH ₂ ⁻
C ₆ H ₆	ca. 43	C ₆ H ₅ ⁻
CH ₄	ca. 48	CH ₃ ⁻



④ 1/5 character

⑤ Inductive/Field effect increases the acidity



Acid	Conjugate base	pK_a	Comments
		~50	charge is localized on one carbon—difficult since carbon is not very electronegative
		~43	charge is delocalized over π system—better but still not really good
		13.5	charge is delocalized over π system but is mainly on the electronegative oxygen—much better
		5	charge delocalized over π system but mainly over two oxygens—better still
		~48	charge is localized on one carbon—again very unsatisfactory
		10	charge is delocalized but mainly on oxygens of nitro group
		4	charge can be delocalized over two nitro groups—more stable anion
		0	charge can be delocalized over three nitro groups—very stable anion

